

One Operator Is Not Enough, Yet: Cost, Learnability, and Guidance Collapse in a Single-Operator Representation of Mathematics at Fixed Scale

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Abstract

One binary operator can, in principle, represent all of elementary mathematics: Odrzywółek (Odrzywółek, 2026) shows every elementary function is constructible from the constant 1 and $\text{eml}(x, y) = \exp(x) - \ln(y)$, turning any expression into a strict binary tree whose internal nodes all carry one operation. Such a single-operator representation is maximally regular—a reason to consider it for graph learning—but its evident cost is size. We measure that cost exactly before asking whether the regularity helps a model, and we do so at one fixed scale: a deterministic, quality-audited corpus of 250,000 expressions with frozen splits. Compiling the corpus with the pinned official compiler yields a median tree expansion factor $\alpha = 40.66$ over the source AST (mean 952.1, p99 10,448.6), and 0/250,000 expressions pass the preregistered per-family thresholds (1.29–1.50, against a whole-grammar counting break-even of ≈ 1.56). Lossless DAG sharing compresses pure-EML trees $39.4\times$ on average, yet the shared EML DAG remains larger than the *uncompressed* AST tree for every one of the 250,000 expressions, the best remaining ratio being exactly $8/7$. Bounded, verified equality saturation improves 23.9–27.6% of costed expressions at 2.4–2.8% mean relative savings, conditional on 60.7% vocabulary coverage. We publish two preregistered nulls (a learned motif vocabulary loses to an equal-budget frequent baseline; a neural cost ranker loses to a structural heuristic) and one preregistered gate verdict, a *fail*, reported with content hashes and not softened. We then release a frozen, leakage-controlled learning apparatus and report its first authenticated verdicts. The single-operator EML-DAG representation is *trainable*: on binary equivalence, two of three seeds reach a validation loss of ≈ 0.53 BCE, below the $\ln 2 \approx 0.693$ chance floor, despite the $40\times$ expansion

(the third seed diverges, at 2.39). Yet a goal-conditioned value head *collapses*: it reaches a low validation MAE (≈ 0.83) while ranking at chance (test-IID Spearman ≈ 0) and emitting a *constant* out-of-distribution predictor, a guidance-signal failure that the rank and OOD diagnostics expose but MAE alone would hide. The study estimates no scaling laws, and frontier LLMs appear only as external context.

1 Introduction

Is one operator enough to represent mathematics for a learning model? A single-operator representation is maximally regular: a graph network reading a pure-EML term never has to learn *what* an operator is, only *where* things connect. The objection is size. Lowering even $\sin(x)$ to pure EML produces hundreds of nodes, and the standard reply is that sharing (DAGs), macros, and semantic rewriting could close the gap. Whether they do is an empirical question that, to our knowledge, had not been measured exactly at a nontrivial scale—and whether whatever gap remains buys anything as inductive bias is a second question the measurement alone cannot settle. We answer the first exactly and take the first authenticated steps on the second: the cost is disqualifying and unrecovered, the representation is nonetheless trainable, and the guidance signal we train on it collapses.

This paper is deliberately narrow: a *structural feasibility study* under preregistered thresholds and complete denominators that publishes null and failed outcomes alongside positives, separating what the representation *costs* structurally (measured in full at production scale) from whether that cost *buys* anything for learning (which we do not fully answer). We fit no scaling law, run no leaderboard, and use no frontier LLM as a controlled

baseline; every number below matches a committed per-goal summary exactly.

Contributions.

- An exact measurement of the single-operator representation’s structural cost at production (250,000-expression) scale: median expansion $\alpha = 40.66$, with 0/250,000 expressions below the preregistered per-family thresholds (C1–C3, §6).
- Two preregistered nulls and one published gate verdict, a **fail** (Gate G10), reported with content hashes and not softened (C4/C9, §5.1 and §6).
- The first authenticated learning verdicts for the representation: a *trainable* single-operator encoder (equivalence validation loss ≈ 0.53 BCE, below the $\ln 2$ chance floor) alongside a *diagnosed* value-head collapse (low MAE, chance-level ranking) (C5/C7, §6.5).
- A transferable methodological lesson—regression error is not ranking utility for a learned value function—and a frozen, verifier-gated, leakage-controlled apparatus released for reuse (§§3–4).

2 Representations, immutable data, and hypotheses

2.1 Representations

Every representation we compare derives from the same source expression, so differences between them are differences of encoding rather than of content. The structural study (§6) uses four: the ordered source AST as a tree and as a hash-consed DAG (Ershov, 1958; Filiâtre and Conchon, 2006), and the official-v4 pure-EML term as a tree and as a hash-consed DAG. Writing α for the ratio of exact pure-EML tree nodes to source AST nodes, the tree pair fixes α and the DAG pair fixes what maximal sharing recovers from it.

The learning grid consumes a different and smaller set, because two of the structural objects are too large to encode at production scale. Four graph channels enter the six-arm equivalence grid (§4): the AST DAG, the pure-EML DAG, a macro-derived frequent-motif DAG, and a train-only motif-AST fair control. Two

non-graph controls complete the grid: a prefix token sequence over the same expressions, and a floor of 19 predeclared surface counts behind a single linear layer. The floor exists to measure how much of the task surface counting already solves, and its feature list is frozen in code and asserted by tests, so no reader can mistake it for a weak learned model.

Macro and motif graphs are lossless DAGs over approved source constructions, not pure-EML graphs: each macro node carries `is_pure_eml = false`, its pure-EML expansion cost is separate metadata never reported as macro size or α , and the runner flags whether any structural metric may be compared across channels. All graphs follow the standard term-graph model (Barendregt et al., 1987), with retained repeated references (so $f(x, x)$ carries two child references to one interned node) and a canonical node signature over roots, edge directions and roles, values, and family; structural identity is hash equality on that signature. Semantic equivalence is a separate relation, established only by the verifier, and we mark each use.

2.2 Fixed data foundation (C1)

The corpus contains 250,000 expressions, unique under the canonical (`domain_mode`, `sympy_srepr`) identity rather than up to semantic equivalence, with frozen splits of 175,000 / 25,000 / 25,000 / 25,000 (train / validation / test_iid / test_ood) and six source families under exact quotas: algebraic core (70,000), exp/log (40,000), powers/division/rationals (40,000), trig/hyperbolic (40,000), mixed elementary (35,000), and an OOD-stress family (25,000).¹ Generation is deterministic and QA-gated; manifests carry per-shard checksums, the config hash, the seed, and a clean commit.

The held-out split and the OOD family coincide by construction: all 25,000 OOD-stress rows are assigned to `test_ood`. Their profile is a combined stress over size, depth, variable count, and composition—not a depth-controlled holdout, and we do not report it

¹`docs/goals/goal1/CORPUS_QA.md`;
`docs/goals/goal2/GOAL2_SUMMARY.md`, “Corpus and denominators”; `FINAL_REPORT.md`
`#c1-goals-1-5-corpus` (pre-authentication digest index).

as one. This study is fixed to the immutable v1 foundation; regenerating the Goals 1–5 inputs would invalidate the comparison it exists to support.

2.3 Predeclared outcome families and hypotheses

Outcome families were declared before the production runs and reported separately—*structural* (tree/DAG size and depth, pure-EML α , macro size, dictionary-inclusive motif description length (Rissanen, 1978)), *predictive* (the six-arm equivalence grid), *rewrite/proof*, *SR*, and *grammar-v2 conformance*. No pooled score exists across families or across channels within the structural family: the runner names every structural metric per channel and the plotting code refuses a shared α column.

Three preregistered hypotheses motivate the structural program. **H1**: raw pure-EML expansion stays below the per-family counting thresholds $\alpha_{\text{tree}} < 1 + \ln K / \ln 4L$ (bounded vocabulary K , literal bound L ; range 1.291–1.500). **H2**: exact structural sharing closes whatever gap H1 leaves, so a shared pure-EML DAG reaches parity with the uncompressed AST tree. **H3**: verified semantic rewriting and learned motif vocabularies close the residual. Structural statistics use all 250,000 rows; semantic auditing draws on a separate bounded sample and we never extrapolate it (§8).

3 Compact models and fair compute

This section and §4 describe committed, CI-tested apparatus; no number in them is a result (the learning verdicts are in §6). We describe it because its frozen design, not its output, is a contribution.

3.1 A compact GINE-style graph encoder

The encoder is a compact variant of GINE (Xu et al., 2019; Hu et al., 2020): three layers, hidden width 64, and about 0.2M parameters. It uses residual connections, GraphNorm or LayerNorm, dropout, a two-layer FFN, and sum pooling. For a message edge $u \rightarrow v$ the edge feature and update are

$$a_{uv} = E_{\text{dir}}(d_{uv}) + E_{\text{slot}}(s_{uv}) + E_{\text{role}}(r_{uv}), \quad (1)$$

$$\begin{aligned} \tilde{h}_v^{(\ell+1)} &= \text{MLP}_\ell \left((1 + \epsilon_\ell) h_v^{(\ell)} + m_v^{(\ell)} \right), \\ m_v^{(\ell)} &= \sum_{u \in \mathcal{N}(v)} \text{ReLU} \left(h_u^{(\ell)} + W_\ell a_{uv} \right), \end{aligned} \quad (2)$$

with the ordering fixed by the frozen implementation. The four arms differ only in their input (AST-DAG, pure-EML-DAG, frequent-motif DAG, motif-AST). Every compressed graph is forced to reconstruct.

3.2 Task composition and controls

For symmetric equivalence classification, the shared endpoint encoders produce graph embeddings g_a, g_b , which can be combined as

$$p(a, b) = [g_a + g_b, |g_a - g_b|, g_a \odot g_b], \quad (3)$$

symmetric in a, b by construction. The transparent floor uses 19 predeclared surface counts that every learned arm must beat: operator, primitive, variable, size, and depth counts. Vocabulary embeddings count toward the matched parameter budget, since pure-EML label vocabularies differ per channel and shift the totals.

3.3 Compute matching and training

Because pure-EML graphs are much larger than AST or macro graphs for the same expression, equal parameters are *not* equal compute. We therefore report, jointly with every quality number, the full compute envelope: parameters, measured and estimated FLOPs, wall time, peak host and device memory, GPU-hours, graph/token size, effective examples and nodes per update, batch and accumulation budgets, optimizer and step budget, early-stopping state, precision, and deterministic settings. Hidden width and the virtual-node flag are frozen only by a measured parameter/FLOP preflight on production vocabularies, which do not exist yet, so the grid config records `pending_preflight_freeze` rather than a guess (Table 1). We use three seeds and publish every seed row; the mean never replaces the raw rows. The default profile is 2×H100 80 GB.

4 Bounded tasks and verification

We evaluate representation quality through bounded, verifier-gated tasks rather than a single aggregate score. Every task has frozen identifiers, declared budgets, retained failure states,

Width	Virt. node	Params	Target
64	no	185,732	below
64	yes	210,692	within
96	no	343,044	within
96	yes	398,916	within

Table 1: Measured encoder parameter counts on the fixture vocabulary (Goal 6 preflight). Width-64 without a virtual node sits marginally under the $\approx 0.2\text{M}$ lower bound; this is recorded, not padded to a round number. Production vocabularies will move these totals, which is exactly why vocabulary embeddings count toward the match. The final freeze is one measured choice, not a tuned one.

and explicit denominators; unavailable cells are reported as missing, not encoded as zero.

4.1 Equivalence

The production equivalence specification fixes 50,000/5,000/5,000/5,000 accepted pair records (train/val/IID/OOD). Each accepted pair is a concrete, replayable rewrite trajectory—rule identifier, direction, occurrence path, bindings, assumptions, source and successor signatures, and separate verifier evidence—and source-, e-class-, and trace-relative grouping prevents leakage across partitions, with rejected, invalid, unsupported, and timeout candidates kept in typed ledgers. The build aligns every endpoint across four graph channels (AST-DAG, strict pure-EML-DAG, a macro-derived frequent-motif DAG, and a train-only AST motif control matched on dictionary-entry count and MDL bits) plus sequence and feature-count controls; IID, OOD-stress, depth-OOD, and family-OOD views never alter the base split totals.

The pipeline has run end-to-end on a byte-identical 10,000-row pilot corpus using the real rule engine and replay verifier, accepting 9,039 records in 34 CPU minutes (6,867 replay-verified positives, 2,172 exact-rationally refuted negatives) and retaining 2,824 rejections and 3,133 typed failures. It is reported honestly as `complete_short` relative to the production target rather than resampled (Table 5); its `test_ood` split holds only one accepted negative because the stress family offers few same-family, size-matched neighbors, an imbalance we retain and report.

4.2 Rewrite-step prediction

Rewrite-step prediction uses one immutable 6-arm $\times$ 3-seed design: the same goal-conditioned GNN over each graph channel, a compute-matched prefix transformer, and a uniform policy sampled from the exact registry-derived legal-action mask. Demonstration-action top- k match is reported separately from exact-successor-structure top- k match, alongside the verifier-valid proposal rate and complete attempted/valid/invalid/unsupported/timeout denominators. Semantic equivalence alone is not counted as demonstration correctness.

4.3 Proof and simplification

The proof benchmark contains exactly 256 frozen problems and the simplification benchmark 1,000 frozen expression identifiers. A proof succeeds only when it reaches the exact target structural signature, its complete transition trace replays, and its terminal claim is verifier-confirmed; lower cost, string agreement, or an unsupported producer claim is not proof success. Uniform, policy, policy-plus-value, and transformer methods share identical beam, expanded-node, proof-depth, wall-time, and verifier-call budgets, and each unsuccessful row is charged at least its expanded-node budget so crashes and timeouts cannot appear artificially efficient. Simplification selects the lowest-cost valid state visited under one preregistered exact `pure_aml:official_v4` cost.

4.4 Symbolic regression

The symbolic-regression benchmark comprises 256 frozen synthetic tasks and a restricted 32-task Feynman-style subset (Udrescu and Tegmark, 2020), with a 100-formula exclusion ledger and verifier-confirmed recovery as the sole criterion; the canonical manifests reproduced identically across three independent CPU builds. The planned comparison includes EML-guided and AST-guided search, a matched PySR baseline or labeled GP fallback, and a compact transformer-SR model under frozen budgets and three seeds, reporting verifier-confirmed recovery over both attempted and verifier-supported denominators alongside the accuracy-complexity Pareto front rather than best accuracy alone.

5 Domain conformance and external context

5.1 Grammar v2 and the published Gate G10 verdict (C9)

The grammar-v2 workstream adds opt-in `asin/acos/atan/π/e` compiler support with the v1 default untouched. Its preregistered bounded conformance audit (74 retained rows, all quotas met, no dropped denominators) was executed on CPU (Apple M1 Pro, Python 3.12.5) at a clean pinned commit and **fails**: zero integrity, coverage, or v1-drift failures, and exactly the eight preregistered `asin/acos` endpoint cells (both compiler modes) nonfinite at 100-digit working precision under the pre-declared 10^{-60} tolerances.² The conformance build produced a byte-identical `records.jsonl` content hash across three executions.³ At 100 digits the tolerances retain 40 guard digits; independent IEEE endpoint probes stay finite and close to the principal references, but because endpoints are required valid source cases the gate correctly fails rather than hiding them as unsupported or lowering precision until they appear finite. We publish the fail; no criteria revision is pre-authorized. Gate G10 is conformance-only: it licenses no corpus-v2, expansion, compression, or learning claim.

5.2 Frontier-LLM panel

An external, verifier-normalized reference panel is implemented but is *external context only*: no row it produces may enter the controlled gates G6–G11. The frozen workload is 100 proof and 100 SR tasks per model across four providers (800 retained rows when a model completes), recording every provider fact—model ID and snapshot, prompts and hashes, budgets, usage, cost, raw responses, and verifier-confirmed correctness. No paid call has been made; the panel ships with paid calls disabled and a zero-cost guard, and claimed correctness is kept distinct from verifier-confirmed correctness.

²`docs/goals/goal10/GOAL10_SUMMARY.md`, “Published verdict”; `FINAL_REPORT.md`
#c9-grammar-v2-conformance.

³Content SHA-256 9649faae3991f8c5
4f8437ac1ab1a9a334606e0499413a04
8084a72e23da80e9; manifest SHA-
256 e25beadd57128a7712bf8edda3d998d5
e708cd55930253d37bfb544947ee6ec2; criteria
SHA-256 83e305e6f1cb76bbd84f5c877a8d3590
4f29484fc4cf95c507ce9a404fd91c62.

6 Results at fixed scale

Table 2 collects every headline structural number; each row names its committed provenance document. All values come from real CPU production runs recorded in those documents; none is estimated. We organize by outcome rather than by a single leaderboard, and give the nulls and the gate fail the space they need to constrain the answer.

6.1 Raw expansion is disqualifying (C2)

Compiling all 250,000 expressions with the pinned official-v4 compiler succeeded on every row and produced a median tree expansion factor $\alpha = 40.6602$ over the source AST (mean 952.1371; full distribution in Appendix A).⁴ The mean/median gap is a heavy right tail dominated by the trig/hyperbolic family (median 270.46), not a typical 952-fold expansion. Against the six preregistered per-family thresholds $1 + \ln K / \ln 4L$ (1.2914–1.5000) and a whole-grammar counting break-even of ≈ 1.56 , the strict pass count is 0/250,000: no expression in any family is close to the regime where a reduced operator vocabulary pays for its node count (Table 3). These thresholds are descriptive combinatorial references, not performance laws.

6.2 Exact compression is not competitiveness (C3)

Lossless hash-consed sharing compresses expanded EML trees by a mean factor of 39.3750, yet the shared EML DAG remains larger than the *uncompressed* AST tree for every one of the 250,000 expressions; the best remaining ratio is exactly 8/7.⁵ The most compressible expression shrinks by 28,087,277/180 and still ends 540/13 times larger than its AST (mean EML-DAG/AST-tree 8.3344, EML-DAG/AST-DAG 10.4750). “Compresses well” and “becomes structurally competitive” are different claims, and only the first is true here.

⁴`docs/goals/goal2/GOAL2_SUMMARY.md`, “Result” and “Expansion distribution”; `FINAL_REPORT.md`
#c2-pure-eml-expansion.

⁵`docs/goals/goal3/GOAL3_SUMMARY.md`, “Structural verdict”; `FINAL_REPORT.md`
#c3-dag-sharing-and-graph-costs.

Quantity	Value	Provenance
Rows processed / exact-count successes (C1)	250,000 / 250,000	Goal 2
Median tree α (EML tree / AST tree) (C2)	40.6602	Goal 2
Mean tree α	952.1371 ($\sigma = 23,155.56$)	Goal 2
p90 / p99 / max α	385.07 / 10,448.60 / 6,481,679.31	Goal 2
Preregistered per-family thresholds 1.2914–1.5000: strict passes	0 / 250,000	Goal 2
Worst family median α (trig/hyperbolic, 40,000 rows)	270.4602	Goal 2
Mean EML self-compression, tree $\rightarrow$ DAG (C3)	39.3750 $\times$	Goal 3
Mean EML DAG / AST tree	8.3344	Goal 3
Mean EML DAG / AST DAG	10.4750	Goal 3
Expressions with EML DAG $\leq$ AST tree	0 / 250,000 (best exactly 8/7)	Goal 3
Largest single self-compression (remaining ratio)	28,087,277/180 (540/13)	Goal 3
E-graph cost coverage (C4)	18,210 / 30,000 (60.700%)	Goal 4
Improved / costed, safe_real	4,349 / 18,210 (23.882%)	Goal 4
Improved / costed, positive_real_formal	5,026 / 18,210 (27.600%)	Goal 4
Mean relative savings over costed rows (incl. unchanged)	2.361% / 2.849%	Goal 4
Unsupported-operator / validation-failure / timeout rows per mode	11,566 / 224 / 0	Goal 4
Null: learned motif vocab. vs. equal-budget frequent (MDL bits, TEST_IID)	324,485,346 vs. 317,678,264	Goal 5
Null: neural ranker vs. structural heuristic (exact-best of 18,050)	8,349 vs. 8,796	Goal 5
Gate G10 conformance verdict (C9)	fail (8 preregistered cells; 74 retained rows)	Goal 10

Table 2: All measured structural results at production scale. Bold marks the winning side or the decisive count. Provenance: committed summaries docs/goals/goal{2,3,4,5,10}/GOAL{2,3,4,5,10}_SUMMARY.md; artifacts are stored off-repository with recorded SHA-256 digests (e.g. Goal 3 run manifest 279b1d01..., analysis fingerprint reproduced twice; Goal 4 run ID 9c26ec30..., rows digest f8fd2e6d...). The committed FINAL_REPORT.md C1–C4/C9 index is a pre-authentication digest index, explicitly not yet final-report-authenticated.

Family	Rows	Median α	Thr.
algebraic core	70,000	25.5778	1.3010
exp/log	40,000	32.7872	1.3382
powers/div/rationals	40,000	44.1579	1.2914
OOD stress	25,000	45.7377	1.3382
mixed elementary	35,000	133.9289	1.4292
trig/hyperbolic	40,000	270.4602	1.5000

Table 3: Per-family median expansion α against the preregistered strict threshold. Every family misses by one to two orders of magnitude; the strict pass count is 0 in each (0/250,000 overall).

6.3 Verified semantic rewriting helps a little, conditionally (C4)

On a frozen, stratified 30,000-expression selection, bounded equality saturation with complete application provenance and independent candidate verification improves 4,349/18,210 costed rows (23.882%) under branch-insensitive real rewrites (**safe_real**) and 5,026/18,210

(27.600%) with guarded positive-real rules (**positive_real_formal**), at mean relative savings of 2.361% and 2.849% over costed rows and a largest single reduction of 302 exact EML-DAG nodes.⁶ Coverage is the binding caveat: 11,566 rows per mode (38.553%) contain operators outside the closed rewrite vocabulary (including *all* trig/hyperbolic and mixed-elementary rows), so these results hold on 60.700% cost coverage and are conditional on it. Each mode retains 224 validation failures (0.747%) and zero timeouts; the source form is always a costed safety anchor, so improvement rates cannot be inflated by dropped rows.

⁶docs/goals/goal4/GOAL4_SUMMARY.md, “Production result”; FINAL_REPORT.md #c4-e-graph-and-motif-compression.

6.4 Two preregistered nulls (C4)

Both Goal 5 comparisons were published in the frozen null-before-positive order.⁷ First, a learned motif vocabulary did *not* beat the equal-budget frequent-motif vocabulary on TEST_IID dictionary-inclusive MDL: 324,485,346 vs. 317,678,264 bits over 25,000 expressions (it did beat the uncompressed macro baseline, 324,485,346 vs. 572,524,716, and the median of 30 fixed random vocabularies, 324,485,346 vs. 368,943,015). Second, a neural cost ranker did *not* beat every structural heuristic under the frozen comparison order: 8,349/18,050 exact-best rankings vs. 8,796/18,050 for the estimated-EML-tree-cost heuristic (and 8,661/18,050 for the AST-DAG-cost heuristic). On these tasks the learned components add nothing over honest structural baselines: the compression does not make the representation competitive, and the learned machinery does not beat the transparent rules it was tested against.

6.5 Learning tracks: a trainable representation, a collapsed guidance signal

The frozen apparatus has now produced its first authenticated verdicts on real GPUs. Of the five tracks, two are measured (equivalence and the value head, below), one is partial—rewrite-step at 13/18 logical cells (5 invalid and retained; separate retrieval grid 15/15, zero authentication errors)—and symbolic regression and the Pareto/Gate-G11 synthesis were not run and are scoped as future work. The two measured tracks pull in opposite directions, and that tension is the finding.

The equivalence track is a viability result. Trained purely on single-operator EML-DAG source cells (three seeds, no AST comparison arm), two seeds reach validation loss 0.5267 and 0.5387 BCE—below the $\ln 2 \approx 0.693$ chance floor for binary equivalence—while the third diverges to 2.3914. That a 40 $\times$ -expanded, maximally regular graph is learnable at all is evidence that the representation’s homogeneity is not by itself an obstacle to a graph model. It is not evidence that EML *helps*: the discriminative six-arm EML-vs-AST grid was not run,

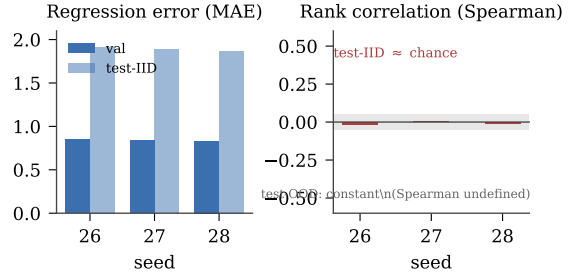


Figure 1: The value-head collapse—the diagnostic MAE alone would miss. **Left:** regression error is unremarkable (validation MAE 0.847/0.838/0.830, test-IID MAE 1.92/1.89/1.87 across seeds 26/27/28). **Right:** the same model ranks at chance (test-IID Spearman $-0.014/0.002/-0.009$, ≈ 0 , grey band), while the out-of-distribution predictor is *constant* and its Spearman is undefined. Low error, no ranking.

and the encoder width and virtual-node choice stayed frozen at `pending_preflight_freeze`. The viability question is answered; the comparison stays open.

The value-head track is a diagnostic result, and a cautionary one (Figure 1). Under a reduced budget (1,500 optimizer steps, three seeds, 452,820 parameters, family `exp_log` held out), the goal-conditioned head reaches a validation MAE of 0.830–0.847 and a test-IID MAE near 1.87–1.92. Read alone, that low error looks like a working regressor. The ranking and out-of-distribution diagnostics say otherwise: test-IID Spearman is -0.014 , 0.002 , and -0.009 across seeds—indistinguishable from zero—and the out-of-distribution predictor is *constant*, so its Spearman is undefined. The official review label is `null_or_collapsed_value_head`: a guidance signal that MAE alone would have passed and that only rank and OOD probes catch. We report it as the methodological finding it is, not a working proof-search heuristic.

7 Discussion

The value-head collapse is not a footnote to the structural result; it is a warning for neural-guided proof and symbolic-regression search, where a learned value or cost head is routinely selected and reported by MAE or MSE. Ours reaches a low validation MAE (≈ 0.83) yet ranks its own held-out states at chance (test-IID Spearman ≈ 0) and degenerates to a constant out of distribution: judged by the usual er-

⁷`docs/goals/goal5/GOAL5_SUMMARY.md`, “Learned and neural comparisons”.

ror metric it would have shipped as a working heuristic. The concrete lesson is that regression error is not ranking utility, and that a guidance model must be reported on rank correlation and out-of-distribution behavior, not error alone—the diagnostics we release with the apparatus. The three findings then compose into one account of the single-operator representation: its structural cost is real and unrecovered by sharing or rewriting, the representation is nonetheless trainable on binary equivalence, but the guidance signal it was meant to supply collapses under bounded compute. Regularity buys learnability without, on this evidence, buying usable guidance—and that gap is where work on single-operator representations should look next.

8 Limitations, integrity, and reproducibility

One scale. Everything is measured at one fixed 250k-v1 scale; we fit no scaling law and extrapolate to no other corpus, budget, or grammar. The expressions are synthetic and generated, and the downstream benchmarks are bounded, which limits external validity. With three seeds, planned learning effects will be descriptive, not asymptotically significant.

Bounded semantic audit. Numeric semantic verification covers a deterministic sample, not the whole corpus. Across the original Goal 2 audit (273 rows) and a widened pilot audit (2,988 rows), the combined evidence base is 3,261 rows with 6 retained mismatches ($\approx 0.1\%$).⁸ The widened audit also hit one row whose numeric materialization deterministically hung (reproduced twice), itself a correctness-surface finding that motivates a per-row timeout. We claim structural determinism corpus-wide but numeric verification only for the audited sample; no mismatch is relabeled or re-sampled away.

Conditional coverage and equal-compute. The e-graph results hold on 60.7% cost coverage, and the excluded families include trig/hyperbolic, the worst-expansion family, so the semantic-compression numbers are not corpus-wide. Because pure-EML expansion

changes input size, equal parameters are not equal compute in the learning tracks, and we report both sides.

Scope of Gate G10. Gate G10’s published fail bounds only the grammar-v2 surface; it neither enables nor blocks any corpus or learning result and stays unresolved by design. The frontier-LLM panel is external context—proprietary data, nondeterminism, and model drift keep it out of every controlled gate.

Artifacts and regeneration. Production artifacts are stored off-repository; the repository commits their SHA-256 digests, per-goal manifests, exact CPU reproduction commands, and a single frozen lock file verified by a CPU-only CI. Generation reproduces on commodity hardware (the 10,000-row pilot regenerated byte-identically twice, corpus hash `df4f3d74...`), and the 250k stage refuses a machine below a twofold memory margin rather than running degraded. Certification is incomplete and recorded as such: the binding machinery authenticated zero artifacts in the archive’s absence and committed a per-claim pre-authentication digest index (27 of 28 manifest entries missing) rather than claiming certification it lacks.

9 Conclusion

At one fixed 250,000-expression scale, the single-operator representation is costly, trainable, and unable to guide. Its structural cost is real and unrecovered—median $40.66\times$ expansion, every one of the 250,000 shared EML DAGs still larger than its *uncompressed* AST tree, 2.4–2.8% recovered by rewriting on 60.7% coverage, and two nulls and a Gate G10 *fail* left unsoftened. Yet the representation is *trainable*: two of three equivalence seeds learn below the $\ln 2$ chance floor despite the $40\times$ expansion. And the guidance signal *collapses*: a value head with low MAE (≈ 0.83) ranks at chance and goes constant out of distribution—a failure MAE alone would hide. The contribution is honest measurement, including where the signal fails, reported without resampling or post-hoc adjustment.

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A Distribution of EML Tree Expansion Factors

Table 4 summarizes the distribution of the tree expansion factor α across all 250,000 expressions in the Goal 2 production run (clean commit 4a5e8b9e..., eight workers, 2,557.26 s wall time). The largest observed expansion occurred for an expression with 13 AST nodes, which compiled to 84,261,831 EML nodes, corresponding to $\alpha = 6,481,679.31$.

B Pilot pair build

Table 5 summarizes the number of sample pairs generated for each split during the 34-minute pilot generation run. The run used the same byte-identical pilot corpus as the final production target (configuration sha256:377d983b..., status complete_short). Each positive pair is accompanied by a verifier trace record that can be replayed to confirm the result.

Negative pairs are instead validated through formal refutations at rationally precise probe

Statistic	Tree α
minimum	1.5000
p10	21.0338
p25	26.3800
median	40.6602
p75	75.2550
p90	385.0723
p95	1,187.8240
p99	10,448.5978
maximum	6,481,679.3077
mean	952.1371
std. dev.	23,155.5594

Table 4: Expansion factor $\alpha = (\text{exact EML tree nodes})/(\text{source AST nodes})$ over all 250,000 rows (Goal 2).

points. The generated split sizes differ from the original target distribution of 50,000/5,000/5,000/5,000 sample pairs, and these differences are reported directly rather than being adjusted through resampling or augmentation.

Once the positive and negative pairs are converted into a unified graph-channel representation, the final dataset contains 18,078 tensor channels. Each sample pair contributes two graph endpoint representations corresponding to the two elements of the pair.

Split	Acc. (pos+neg)	Rej.	Fail	Short
train	6,281 (4,568+1,713)	1,786	2,432	43,719
validation	893 (657+236)	263	343	4,107
test_iid	864 (642+222)	276	358	4,136
test_ood	1,001 (1,000+1)	499	0	3,999
total	9,039 (6,867+2,172)	2,824	3,133	—

Table 5: Pilot pair build (Goal 6 pipeline). Accepted = positives + negatives; rejected and failed rows are retained with typed reasons. The lone test_ood negative reflects few same-family size-matched neighbors in the stress family and is reported, not patched.